The Effect of Socioeconomic and Food Access Factors on Chronic Health Outcomes in Lancaster County, NE

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Abstract

This study investigates the relationship between food accessibility, socioeconomic factors, and chronic health outcomes—specifically obesity, type II diabetes, and coronary heart disease—within Lancaster County, Nebraska. While traditional "food desert" research emphasizes physical distance to grocery stores as the primary driver of poor health, our analysis suggests a more complex interaction. By synthesizing spatial data from the USDA and OpenStreetMap with health and economic indicators from the CDC and the U.S. Census Bureau, we developed a multiple linear regression model to quantify these relationships at the census tract level.

Our findings indicate that socioeconomic variables, particularly educational attainment and poverty levels, are significantly stronger predictors of chronic health outcomes than physical proximity to food outlets. The regression model achieved an $R^2 \approx 0.49$, suggesting that nearly half of the variation in obesity is explained by socioeconomic status, while distance to grocery stores remained a weak predictor. Spatial mapping further reveals distinct geographic clustering of health risks that aligns more closely with economic disadvantage than with store density. These results suggest that public health interventions should prioritize mitigating financial and educational barriers rather than focusing solely on the physical placement of new grocery stores.

Contents

Chapter 1

Introduction

1.1 Motivation

In recent years, public health discussions have frequently centered on the concept of "food deserts" and "food swamps," areas where physical distance to food sources is believed to be a primary barrier to maintaining a healthy diet. However, this simple geographic explanation often fails to account for the complex reality of how people actually live and access healthy food. In a city like Lincoln, NE—which features a dense urban core alongside sprawling rural and neighborhood outskirts—the relationship between where you live and how healthy you are is rarely straightforward [3].

Using the University of Wisconsin Population Health Institute's *County Health Rankings and Roadmaps* website, Lancaster County, NE scored an 8.1 out of 10 on the *Food Environment Index* (FEI) for 2025, a measure of factors contributing to a healthy food environment [2]. This score is high relative to Nebraska (7.7) and the United States (7.6). However, examining indicators under "Population Health and Well-Being" and "Social and Economic Factors" reveals a more concerning picture:

Food Insecurity: 15%
Physical Inactivity: 21%
Adult Obesity: 36%

This discrepancy motivates our central question: are there hidden drivers of health outcomes—such as obesity, type II diabetes, and coronary heart disease—that correlate with food access conditions but are not captured by the FEI score?

1.1.1 Central Question and Problem

The goal of this research project is to determine which factors are the most powerful predictors of chronic health outcomes. Specifically, we test whether physical distance to a grocery store is the primary barrier, or whether socioeconomic factors—such as income, education, and vehicle access—are the stronger drivers of health disparities. By analyzing Lancaster County at the census tract level, we aim to identify the structural barriers that keep residents from reliably accessing nutritious food.

1.1.2 Importance

Understanding these factors is vital for local policymakers and health organizations. If obesity in Lincoln is driven by proximity to food, then building more stores is the solution.

If it is driven by socioeconomic conditions, the solution may involve improving public transit or expanding economic support for specific communities. This distinction has direct implications for how resources are allocated and which interventions are likely to succeed [4, 3].

1.1.3 Real-World and Mathematical Significance

From a real-world perspective, this project identifies vulnerability hotspots where health risks are highest, allowing policymakers to move beyond the one-size-fits-all definition of food deserts. From a mathematical perspective, the project applies multiple linear regression and random forest modeling to quantify how much of the variation in health outcomes can be explained by socioeconomic and geographic variables. An $R^2 \approx 0.49$ in the preliminary linear model suggests that nearly half of a census tract's health profile can be predicted from a small set of socioeconomic indicators—a meaningful result for applied social science.

1.2 Overview of the Project

This study synthesizes health data from the CDC with socioeconomic data from the U.S. Census Bureau and spatial data from OpenStreetMap. We begin by establishing a mathematical foundation in correlation and multiple linear regression, then apply spatial mapping to visualize patterns across Lancaster County. The report is structured as follows: Chapter 2 details the data and modeling assumptions; Chapter 3 provides mathematical background; Chapter 4 describes the analysis; and Chapter 5 presents our findings and conclusions.

1.2.1 Key Terms and Definitions

1. **Food Insecurity:** Limited or uncertain access to adequate food [3].

2. **Food Desert:** An area that lacks a grocery store, or where consumers must travel significant distances to reach one [12].

3. **Census Tract:** A small, relatively permanent subdivision of a county used by the U.S. Census Bureau for statistical data collection. Census tracts are designed to be fairly homogeneous with respect to population characteristics, economic status, and living conditions [13].

Chapter 2

Data and Assumptions

2.1 Data

This study combines several datasets to examine how socioeconomic conditions and food accessibility relate to chronic health outcomes across census tracts in Lancaster County, Nebraska. The unit of observation throughout the analysis is the census tract.

Socioeconomic data was gathered from the American Community Survey (ACS) through the U.S. Census Bureau [1]. The ACS variables used in this study are percent uninsured, percent of households without access to a vehicle, percent without a high school diploma, and poverty rate. These variables provide tract-level measures of economic vulnerability, educational disadvantage, and transportation limitations that may constrain food access and health outcomes. The ACS table identifiers used were:

1. **S1501** – Educational Attainment (percent without a high school diploma)
2. **S2701** – Health Insurance Coverage (percent uninsured)
3. **B08201** – Household Size by Vehicles Available (percent without a car)
4. **S1701** – Poverty Status in Past 12 Months (poverty rate)

Health outcome data was obtained from the CDC PLACES data portal [8]. The primary health variables used were obesity prevalence, diabetes prevalence, and coronary heart disease prevalence, defined as follows:

1. **Obesity:** Body mass index (BMI) greater than 30.0 kg/m^2 [7].
2. **Type II Diabetes:** A condition in which cells cannot adequately absorb blood sugar due to insulin resistance, preventing normal glucose uptake [9].
3. **Coronary Heart Disease:** Characterized by plaque buildup within the arteries, which narrows arterial walls and can partially or fully restrict blood flow to the heart [6].

These variables serve as the response variables and allow comparison of chronic health outcomes across communities. All three were selected because they share similar dietary and food access risk factors.

All datasets were merged at the census tract level using a common geographic identifier (GEOID). Preprocessing steps included standardizing tract identifier formats across sources, converting string-formatted percentages to numeric values, handling missing values, and constructing derived variables such as fast food density, grocery density, and distance-based access measures.

Food access variables were constructed using geographic data from OpenStreetMap, processed with OSMnx [5]. This provided locations of grocery stores, supermarkets, convenience stores, and fast food establishments in Lancaster County. Distance to the nearest grocery store was calculated from each tract's centroid using straight-line distance in projected coordinates, converted to kilometers. Density variables were computed by spatially joining food locations to census tracts and dividing counts by tract area in square kilometers.

2.2 Assumptions and Limitations

1. Census tracts are treated as independent observations. In practice, neighboring tracts may share similar characteristics, so this is a simplification. However, it enables standard regression and correlation methods at the tract level.

2. The multiple linear regression model assumes an approximately linear relationship between obesity prevalence and the explanatory variables, with error terms of mean zero and roughly constant variance. Residual analysis suggests this is a reasonable first approximation.

3. Geographic distance to grocery stores is used as a proxy for food access. This is a useful but incomplete measure, as true food access also depends on transportation availability, transit coverage, affordability, and individual mobility.

4. Data is aggregated at the census tract level. Tract-level averages may conceal within-neighborhood variation, and conclusions should be interpreted as tract-level patterns rather than individual-level claims.

5. OpenStreetMap food location data is assumed to reasonably represent the local food environment. Some store locations may be missing or inconsistently categorized; to reduce this limitation, a broader set of grocery-related categories was used when constructing food access variables.

6. This study is limited to Lancaster County, Nebraska. The relationships identified here may not generalize to counties with different urban structures, transportation systems, or socioeconomic compositions.

Chapter 3

Mathematical Background

This chapter introduces the mathematical concepts used to analyze the relationships between socioeconomic factors, food access, and health outcomes.

3.1 Correlation

To measure the strength of the linear relationship between two variables, we use the Pearson correlation coefficient [10]. For two variables X and Y, the correlation is defined as

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

The value of r lies in the interval $[-1, 1]$, where $r \approx 1$ indicates a strong positive linear relationship, $r \approx -1$ indicates a strong negative relationship, and $r \approx 0$ indicates little to no linear relationship. In this study, correlation analysis served as an initial screening tool to identify which variables were most strongly associated with obesity before fitting regression models.

3.2 Multiple Linear Regression

To model the relationship between obesity prevalence and multiple explanatory variables simultaneously, we use a multiple linear regression model of the form

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \varepsilon_i,$$

where:

- Y_i is the obesity rate in tract i,
- X_{1i} represents percent without a high school diploma,
- X_{2i} represents distance to the nearest grocery store,
- X_{3i} represents fast food density,
- $\beta_0, \beta_1, \beta_2, \beta_3$ are model coefficients,
- ε_i is an error term assumed to be independent with mean zero and constant variance.

The model produces coefficient estimates describing the direction and relative magnitude of each predictor's association with obesity, providing an interpretable baseline for comparing socioeconomic and geographic factors.

3.3 Coefficient of Determination

The coefficient of determination, R^2, measures the proportion of variance in the response explained by the model:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}.$$

A value close to 1 indicates a strong fit; a value close to 0 indicates weak explanatory power. However, R^2 alone does not confirm that the model is correctly specified.

3.4 Adjusted R^2

Because R^2 increases whenever variables are added to a model, we also consider the adjusted coefficient of determination:

$$R^2_{\text{adj}} = 1 - (1 - R^2)\frac{n-1}{n-p-1},$$

where n is the number of observations and p is the number of predictors. A large gap between R^2 and R^2_{adj} suggests that additional predictors may not contribute meaningful explanatory power.

3.5 Root Mean Squared Error

The root mean squared error (RMSE) measures the average magnitude of prediction error:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}.$$

Lower RMSE indicates better predictive accuracy. Because RMSE is expressed in the same units as the response variable, it is more directly interpretable than R^2 for assessing practical prediction quality.

3.6 Residuals

Residuals measure the difference between observed and predicted values:

$$e_i = y_i - \hat{y}_i.$$

If residuals appear randomly scattered around zero with no systematic patterns, the linear model is considered a reasonable fit. Patterns in the residuals may indicate missing structure or model misspecification.

3.7 K-Fold Cross-Validation

To evaluate model stability, we implemented K-fold cross-validation with $K = 5$. The dataset was partitioned into disjoint subsets $D_1, \ldots, D_K$. For each fold k, the model was trained on $D_{-k} = D \setminus D_k$ and evaluated on D_k. The cross-validated R^2 is

$$R^2_{CV} = \frac{1}{K} \sum_{k=1}^{K} R^2_k.$$

This procedure reduces dependence on any single train-test split and provides a more reliable estimate of out-of-sample performance.

3.8 Random Forest

A random forest is an ensemble of decision trees $\{T_b\}_{b=1}^{B}$, each trained on a bootstrap sample of the data. For regression, predictions are averaged across trees:

$$\hat{y}(x) = \frac{1}{B} \sum_{b=1}^{B} T_b(x).$$

Random forests do not assume a linear relationship between predictors and the response, making them useful as a comparison model. Feature importance values from the random forest measure each variable's relative contribution to predictive accuracy, providing a complementary perspective to linear regression coefficients.

Chapter 4

Analysis

4.1 Overview of Approach

To address the central question, we model obesity prevalence as a function of both socioeconomic variables and food access measures, allowing direct comparison of their relative importance. We proceed through correlation analysis, linear regression, residual and cross-validation evaluation, and a random forest comparison.

4.2 Correlation Analysis

Pairwise correlations between variables reveal a clear divide between socioeconomic and food access predictors. Percent without a high school diploma has the strongest positive correlation with obesity ($r \approx 0.75$), followed by poverty rate and percent uninsured. In contrast, food access variables—grocery density and distance to the nearest store—show weaker and less consistent relationships. These patterns suggest that socioeconomic conditions are more closely associated with health outcomes than physical food proximity.

Figure 4.1 displays the full correlation matrix.

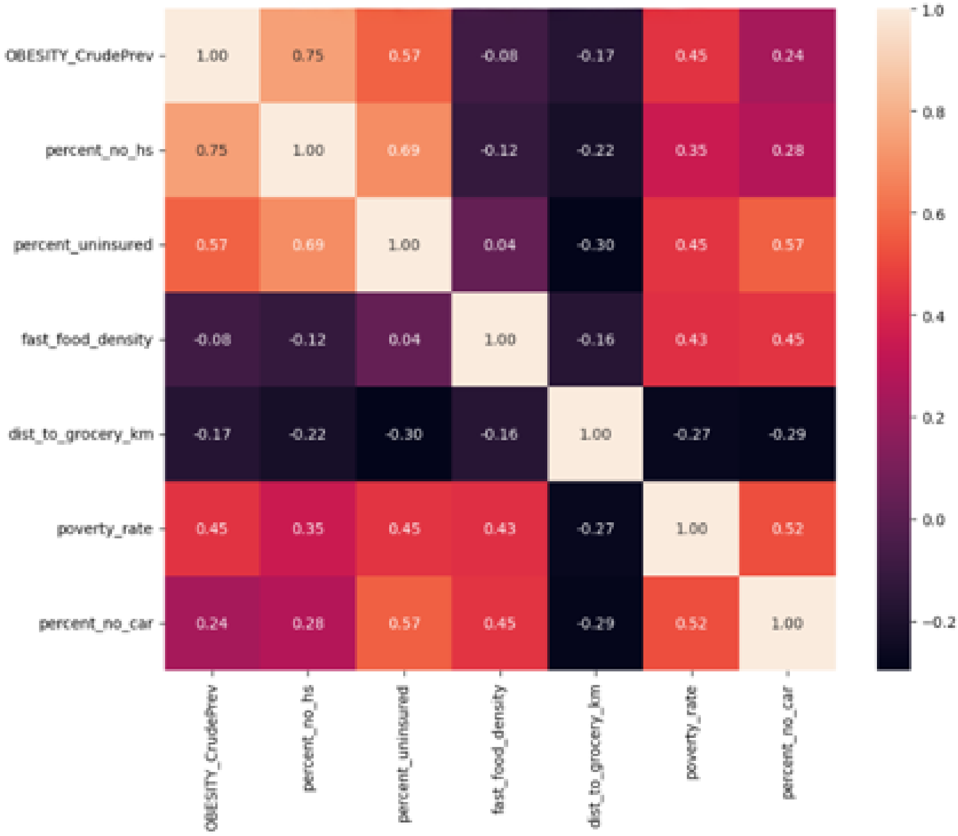

Figure 4.1: Correlation matrix of socioeconomic and food access variables

Figure 4.2 illustrates the strong positive relationship between lack of educational attainment and obesity prevalence.

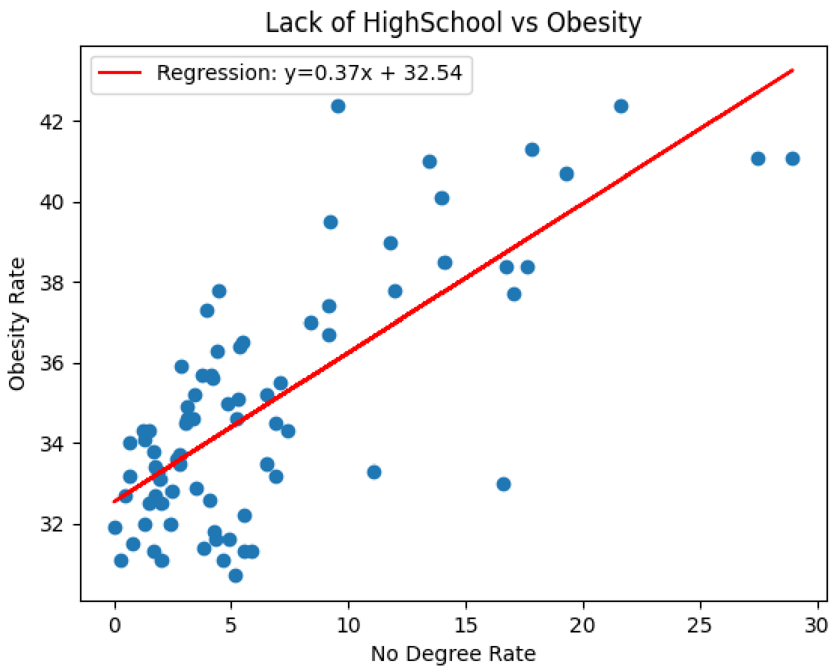

Figure 4.2: Scatter plot with regression line: percent without a high school diploma vs. obesity prevalence

4.2.1 Spatial Distribution of Obesity

Spatial visualization of obesity prevalence across Lancaster County census tracts reveals that health outcomes are not randomly distributed. The highest-vulnerability neighborhoods in the urban core align closely with the highest concentrations of poverty and lowest levels of educational attainment. While the county-wide average obesity rate is 34.9%, specific urban tracts consistently exceed the 40% threshold.

4.2.2 The Geometry vs. Functionality Pivot

Our initial hypothesis treated straight-line distance from a census tract centroid to the nearest grocery store as the primary barrier to food access. However, spatial analysis and regression testing showed this metric to be a poor predictor of health outcomes, yielding an $R^2 \approx 0.04$ and a near-zero correlation of $r = -0.19$.

This finding prompted a necessary methodological reframing. High-obesity tracts appeared across all distance ranges, including tracts immediately adjacent to grocery stores. The core insight is that geometric proximity does not equal functional access. A 0.8-mile grocery run is a minor inconvenience for a resident with a vehicle, but for a resident without a car, the same trip may require a 3.2-mile transit route involving multiple bus transfers and approximately 45 minutes of travel time.

4.2.3 Transit Access and Socioeconomic Factors

To investigate functional access more directly, we overlaid StarTran bus route data onto vulnerability heatmaps, focusing on high-frequency corridors:

- **Route 44 (O Street):** This high-coverage corridor cuts through Lincoln's most vulnerable tracts. Despite strong bus coverage, poor health outcomes persist in these areas, indicating that transit availability alone cannot close the health gap.
- **Route 27 (North 27th St):** This route connects lower-resource neighborhoods to affordable food outlets such as Aldi. While it provides a vital service, its presence does not statistically reduce the high obesity rates observed in these tracts.

These findings confirm that the barriers to healthy outcomes in Lancaster County are not purely spatial or transit-based, but are heavily shaped by financial and educational constraints. To capture this compounding effect, we introduced a **Neighborhood Vulnerability Index** (V):

$$V = \frac{(\text{Fast Food Density}) \times (\text{Housing Cost Burden } \%)}{\text{Distance to Healthy Grocery Store} \times \text{Transit Frequency}}$$

4.2.4 Transit Case Studies

Figure 4.3 provides an overview of how major transit lines intersect with areas of high poverty concentration in Lincoln's urban core.

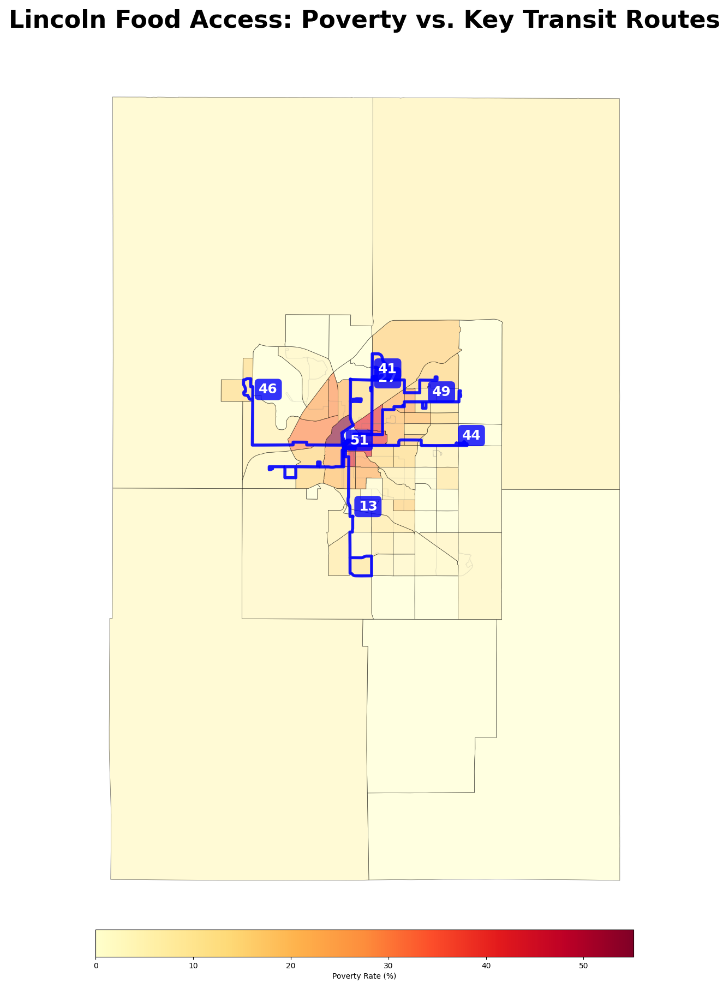

Figure 4.3: Lincoln food access overview: poverty rate by census tract with key StarTran transit routes

Figures 4.4 and 4.5 present case studies of two transit corridors that serve as critical connections for residents in high-vulnerability tracts.

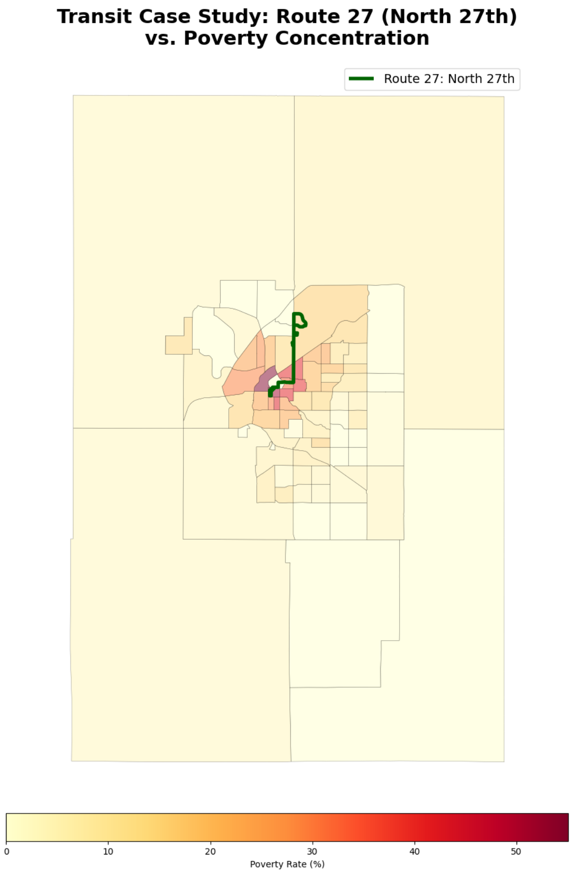

Figure 4.4: Route 27 (North 27th St) case study

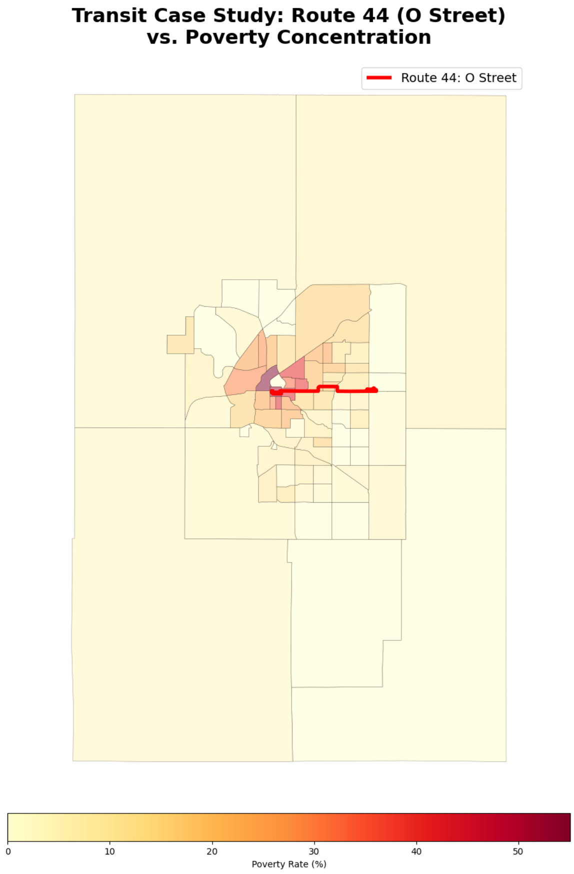

Figure 4.5: Route 44 (O Street) case study

As shown in Figures 4.4 and 4.5, routes such as North 27th and O Street are essential connectors between high-poverty neighborhoods and grocery outlets. However, residents who depend on fixed transit schedules—particularly during off-peak hours or when transporting groceries—continue to face meaningful access friction.

4.2.5 Statistical Validation of Spatial Patterns

The spatial clusters observed in our maps are supported by the statistical models. A multiple linear regression model applied to these tracts achieved $R^2 \approx 0.49$, indicating that socioeconomic constraints explain nearly half of the variation in obesity. Educational attainment emerged as the dominant predictor, with percent without a high school diploma showing a correlation of $r \approx 0.75$ with obesity.

A random forest model improved explanatory power to $R^2 \approx 0.59$. Feature importance analysis confirmed the primacy of education: lack of a high school diploma carried an importance weight of 0.721, far exceeding physical distance to a grocery store at 0.143. The consistency of this finding across both linear and nonlinear models strengthens confidence in the overall conclusion.

Figure 4.6 shows the spatial distribution of obesity across Lancaster County census tracts.

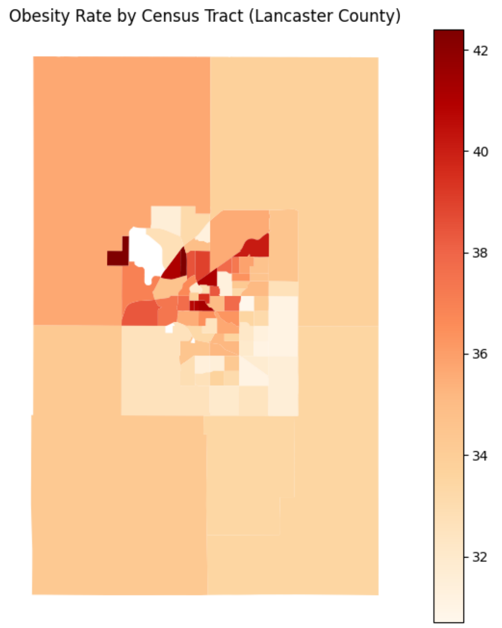

Figure 4.6: Obesity prevalence by census tract in Lancaster County

4.3 Linear Regression Model

We fit a multiple linear regression model of the form

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \varepsilon_i,$$

where the predictors represent percent without a high school diploma, distance to the nearest grocery store, and fast food density. The fitted model achieves $R^2 \approx 0.49$ and RMSE ≈ 0.66. Since obesity is measured as a percentage, the RMSE indicates that predicted values are typically within less than one percentage point of observed values—a relatively close fit on this data split.

Figure 4.7 compares predicted and actual obesity values. The model captures the general trend, but the spread of points and presence of outliers indicate that not all variation is explained.

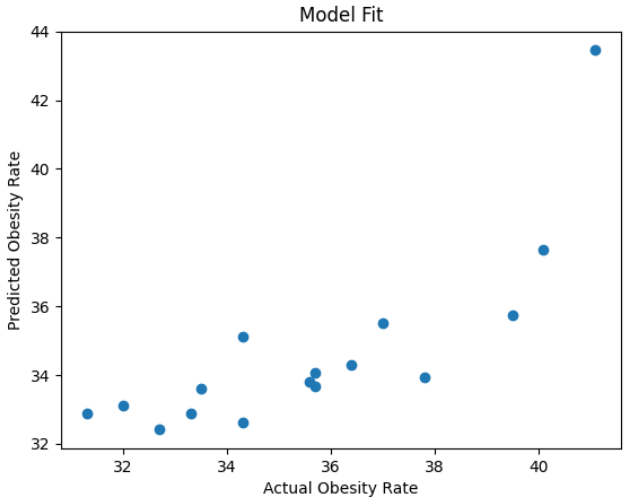

Figure 4.7: Predicted vs. actual obesity values from the linear regression model

4.3.1 Model Considerations and Evaluation

While R^2 measures the proportion of variance explained by the model, it can be misleading if the functional form is misspecified. Adjusted R^2 is also reported to account for model complexity. Correlation analysis revealed moderate associations among some socioeconomic variables, indicating potential multicollinearity; coefficients are therefore interpreted with caution, with greater weight placed on overall model behavior and relative predictor importance rather than individual coefficient significance.

Alternative specifications were also considered. Because obesity prevalence is a proportion, fractional logistic regression could be appropriate in future work. A log transformation of distance may better capture diminishing access effects. The linear model is retained here for interpretability and because it provides a reasonable approximation given the observed data.

4.4 Residual Analysis

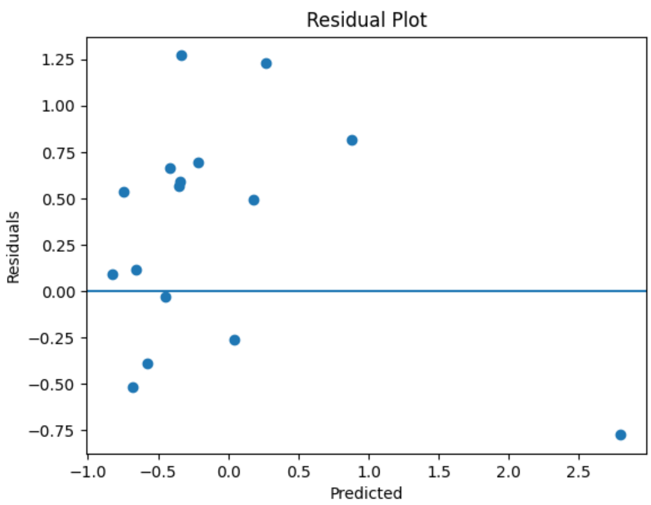

Figure 4.8: Residual plot for the linear regression model

The residuals shown in Figure 4.8 are mostly randomly scattered around zero, suggesting that the linear model provides a reasonable approximation. One notable outlier is visible, indicating that the model does not fully capture variation in all tracts.

4.5 Cross-Validation

Repeated 5-fold cross-validation yields a mean $R^2 \approx 0.33$ with a relatively large standard deviation across folds. This is lower than the single-split result, which is expected given the small number of census tracts and the complexity of health outcomes. The cross-validated result indicates that the model identifies real but imperfect patterns in the data. Predictive stability could likely be improved by incorporating additional variables such as age structure, affordability measures, or more detailed transportation data.

4.6 Random Forest Comparison

A random forest model (100 trees) achieves $R^2 \approx 0.59$, confirming that nonlinear structure is present. Despite this improvement, the relative importance of predictors remains consistent with the linear model. As shown in Table 4.1, percent without a high school diploma dominates in both models, while food access variables contribute comparatively little.

Feature	Linear Regression Coefficient	Random Forest Importance
Percent No High School	0.784	0.721
Distance to Grocery (km)	0.024	0.143
Fast Food Density	0.014	0.136

Table 4.1: Comparison of predictor influence across linear regression and random forest models

Note that regression coefficients and random forest importance values are not directly comparable, as they measure different aspects of predictor influence. Both, however, identify educational attainment as the dominant factor. We also tested poverty rate and percent uninsured as alternative socioeconomic predictors; while exact rankings shifted slightly, the broad pattern—socioeconomic variables outperforming food access measures—remained consistent across all specifications.

Chapter 5

Results and Conclusions

5.1 Correlation Patterns

The correlation matrix reveals a clear separation between socioeconomic and food access predictors. Percent without a high school diploma shows the strongest positive relationship with obesity ($r \approx 0.75$). Poverty rate and percent uninsured also display moderate positive correlations. In contrast, distance to grocery stores and fast food density show weaker and less consistent relationships, suggesting that physical food access does not strongly explain variation in health outcomes at the census tract level.

5.2 Regression Findings

The multiple linear regression model explains approximately half of the variation in obesity rates ($R^2 \approx 0.49$), with an RMSE of approximately 0.66 percentage points. Percent without a high school diploma carries a substantially larger coefficient than food access variables, confirming that education level is the stronger predictor.

5.3 Model Stability

Cross-validation results show a mean $R^2 \approx 0.33$ with meaningful variance across folds, indicating that the model captures real relationships but is sensitive to data partitioning. Residual analysis supports the use of the linear form as a reasonable approximation, with one notable outlier suggesting the model does not fully account for all tracts.

5.4 Comparison of Models

The random forest achieves higher explanatory power ($R^2 \approx 0.59$) but arrives at the same substantive conclusion: educational attainment accounts for the majority of predictive importance (0.721), while physical distance to grocery stores contributes far less (0.143). Agreement across modeling approaches strengthens confidence in the finding.

5.5 Key Takeaways and Implications

Socioeconomic conditions, particularly educational attainment, are more influential predictors of chronic health outcomes in Lancaster County than physical proximity to food. Even in areas with reasonable geographic food access, underlying economic and educational barriers appear to limit residents' ability to benefit from nearby resources.

This has direct implications for public health policy. Interventions focused solely on placing new grocery stores are unlikely to produce meaningful improvements unless they also address the structural conditions—income, education, transportation—that shape residents' actual ability to access and use those resources.

5.5.1 Spatial Context: Transportation and Food Access

Figure 5.1 overlays bus routes on a heatmap of households without vehicle access, providing spatial context for interpreting food accessibility beyond simple distance.

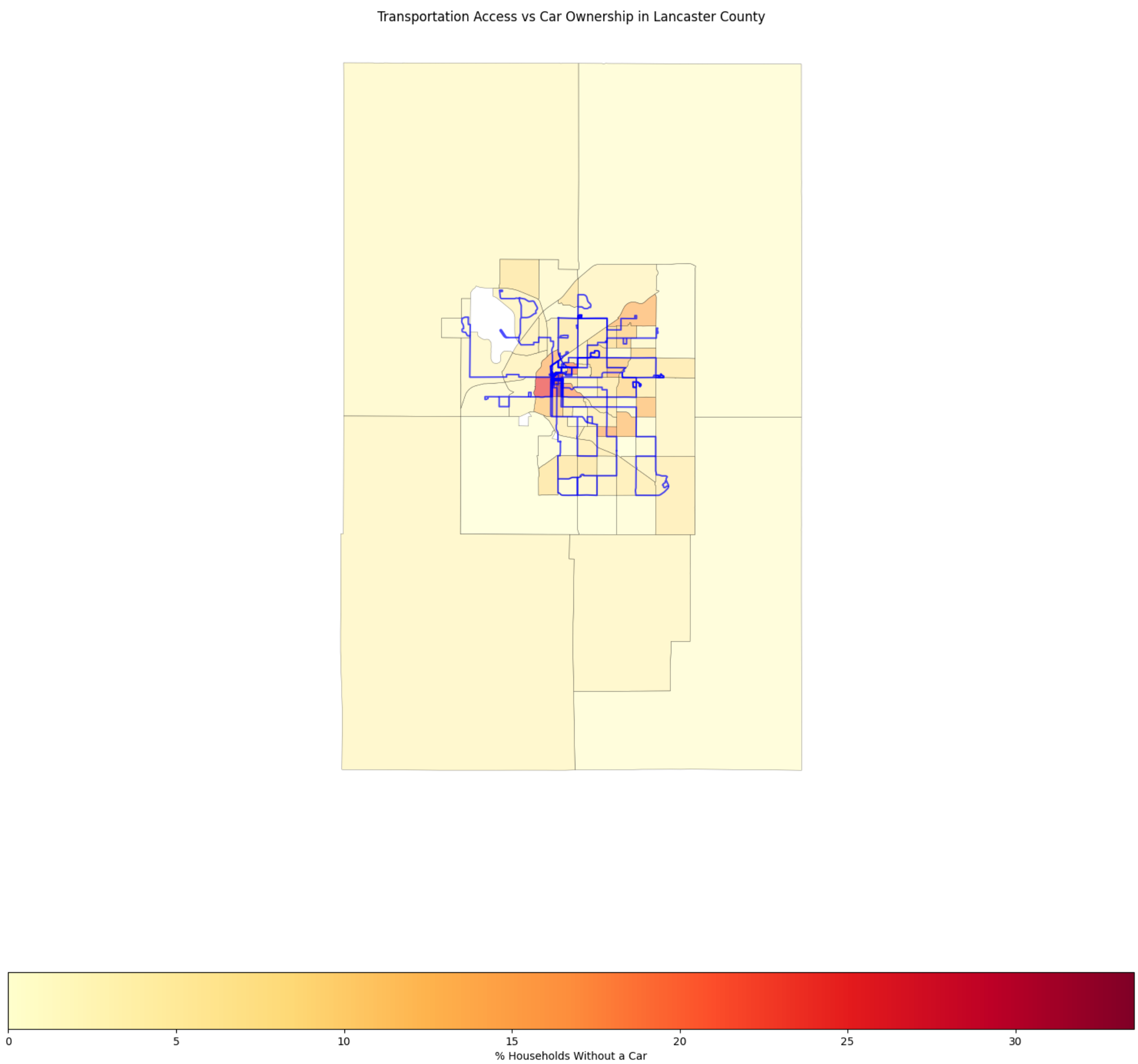

Figure 5.1: Bus routes overlaid on percent of households without a car, by census tract

Many low car-ownership areas are served by public transit routes, particularly in the densely populated urban core. This suggests that some residents can physically reach grocery stores even without a vehicle. However, the persistence of poor health outcomes in these same areas indicates that transit availability alone is insufficient to eliminate access barriers—reinforcing the conclusion that socioeconomic factors play the dominant role.

5.6 Future Directions

Prior research on obesity in Nebraska suggests that age-related and child-focused variables may improve the model's explanatory power [11]. Variables such as SNAP participation, free and reduced lunch eligibility, school meal program access, and the percentage of children in single-parent households could provide a more detailed picture of how socioeconomic conditions affect different populations. Future work could incorporate these variables alongside broader demographic factors such as age distribution, enabling more targeted and effective public health strategies for both urban and rural communities.

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